

# Metasploitable2: UnrealIRCd 3.2.8.1 Vulnerability Assessment

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## 1. Objective

The objective of this assessment was to identify and validate a known vulnerability present in the UnrealIRCd service running on the Metasploitable2 virtual machine. The exercise focused on service enumeration, vulnerability identification, controlled validation within an isolated laboratory environment, and analysis of the security implications associated with the affected service.

## 2. Lab Environment

Attacker Machine: Kali Linux  
Target Machine: Metasploitable2  
Hypervisor: Oracle VirtualBox  
Network Type: Host-Only Network  
Assessment Type: Controlled Security Lab  
Target Service: IRC (Internet Relay Chat)  
Service Port: TCP 6667

## 3. Reconnaissance

Initial enumeration identified an IRC service exposed on TCP port 6667. Service identification indicated the use of UnrealIRCd, a version associated with a historically documented backdoor vulnerability.

## 4. Vulnerability Overview

UnrealIRCd 3.2.8.1 Backdoor Vulnerability (CVE-2010-2075). In 2010, a compromised version of UnrealIRCd was distributed through official project infrastructure. The modified source code introduced a backdoor capable of enabling unauthorized command execution on affected systems.

## 5. Validation Method

The vulnerability was validated in a controlled laboratory environment using the Metasploit Framework running on Kali Linux. Validation confirmed that the target system was susceptible to the known UnrealIRCd backdoor issue.

## 6. Findings

The target IRC service was running a vulnerable build of UnrealIRCd associated with a documented backdoor vulnerability. Successful validation demonstrated that the service could allow unauthorized access to the target system, resulting in complete compromise of the affected host.

## **7. Security Impact**

Potential impacts include unauthorized command execution, unauthorized access, data exposure, persistence mechanisms, and use of the compromised host as a pivot point for additional attacks.

## **8. Remediation**

Replace vulnerable UnrealIRCd versions with trusted releases, validate software integrity, restrict unnecessary exposure, apply security updates, monitor services, conduct assessments, and implement network segmentation.

## **9. Lessons Learned**

Service enumeration is critical. Version identification can reveal known vulnerabilities. Supply-chain compromises present significant risks. Isolated laboratory environments provide safe learning opportunities. Patch management reduces exposure.

## **10. Conclusion**

The assessment confirmed the presence of a known UnrealIRCd vulnerability on the Metasploitable2 target system. The exercise demonstrated the importance of identifying outdated network services and understanding risks associated with compromised software distributions.